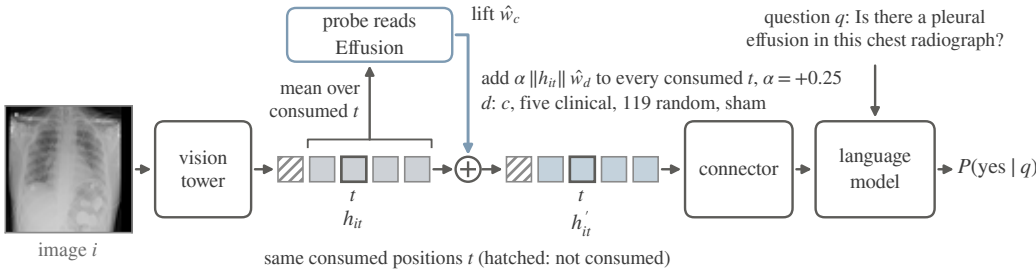


(a) read and write at the final visual block the language model consumes



(b) the grade: one write, three comparisons

$W_{q,d}$ : mean change in  $P(\text{yes} | q)$  under the write of  $d$

- own effect  
 $W_{q,q} > 0$  ✓
- random and sham  
 $W_{q,q} > \text{random p95 and } > \text{sham}$  ✓
- five clinical directions  
 $W_{q,q} - W_{q,d} > 0$  for every  $d \neq q$  ✗

owned: all three hold under simultaneous max- $T$  bounds  
seed cell: stronger competitor

(c) seed cell: Effusion, Qwen2.5-VL-7B, NIH

